LAB 1 – REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

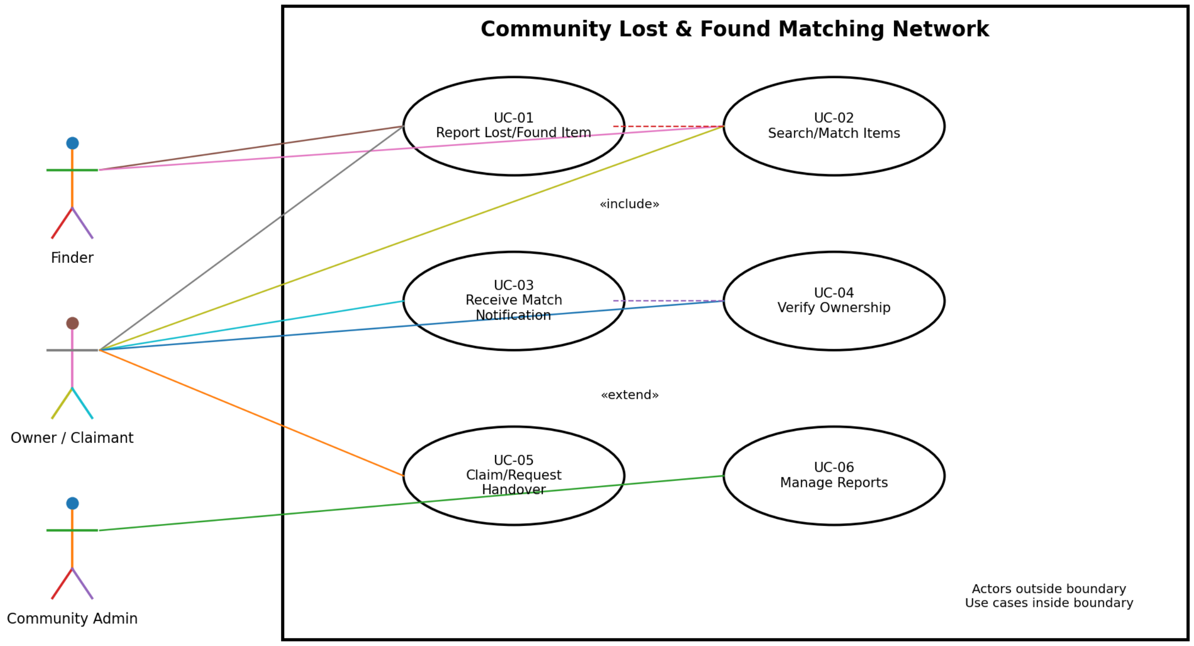
Problem Statement #59: Community Lost & Found Matching Network

Name: Divyanshu Jha **SRN:** PES2UG24CS915

1. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall match newly logged Found Item posts with active Lo[illegible]	High	Pass: A match [illegible]	[illegible]
FR-002	Functional	The system shall allow a user to create and submit a Lost Item or Found [illegible]	High	Pass: A report [illegible]	[illegible]
FR-003	Functional	The system shall notify potential owners when a Found Item [illegible]	High	Pass: A notification is generated within 30 seconds of a successful [illegible]	[illegible]
FR-004	Functional	The system shall allow a claimant to answer private proof-of-ownership [illegible]	High	Pass: [illegible] proceed. Fail: In[illegible]	[illegible]
FR-005	Functional	The system shall allow the Community Admin to review, remove, or flag [illegible]	Medium	Pass: [illegible] a selected report and its [illegible]	[illegible]
NFR-001	Nonfunctional — Performance [illegible]	[illegible] must answer private security questions set by the finder be[illegible]	High	Pass: [illegible] required security [illegible]	[illegible]
NFR-002	Nonfunctional — Performance	The system shall calculate item matches and display matching results [illegible]	High	Pass: At least 95 [illegible] within 5 seconds. Fail: Fewer than 95 [illegible]	[illegible]

2. UML Use-Case Diagram



3. Use-Case Flow — Verify Ownership and Claim Found Item

Actors

Owner / Claimant, Finder

Preconditions

A Found Item report exists; the claimant has received a potential match notification; the finder has configured private proof-of-ownership questions.

Postconditions

The verification result is recorded. If successful, the claimant can proceed with handover; if unsuccessful, contact details remain hidden.

Main Success Scenario

1. Owner/Claimant opens the matching Found Item notification.
2. System displays item details and verification option.
3. Claimant selects Verify Ownership.
4. System displays private security questions.
5. Claimant enters answers.
6. System validates the answers.
7. System confirms successful verification.
8. System allows a handover request.
9. System reveals appropriate handover contact details.
10. Use case ends successfully.

Alternate Flow — Incorrect Security Answer

6a. System informs the claimant that verification failed.
6a1. Contact details remain hidden.
6a2. Claimant may retry.
6a3. After the maximum attempts, the verification request is blocked.
6a4. Use case ends unsuccessfully.